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Review Article

Epidural injections in prevention of surgery for spinal pain: systematic review and meta-analysis of randomized controlled trials

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Abstract

BACKGROUND CONTEXT: Low back pain is debilitating and costly, especially for patients not responding to conservative therapy and requiring surgery.

PURPOSE: Our objective was to determine whether epidural steroid injections (ESI) have a surgery-sparing effect in patients with spinal pain.

STUDY DESIGN/SETTING: The study design was based on a systematic review and meta-analysis.

METHODS: Databases searched included Cochrane, PubMed, and EMBASE. The primary analysis evaluated randomized controlled trials (RCTs) in which treatment groups received ESI and control groups underwent control injections. Secondary analyses involved RCTs comparing surgery with ESI, and subgroup analyses of trials comparing surgery with conservative treatment in which the operative disposition of subjects who received ESI were evaluated.

RESULTS: Of the 26 total studies included, only those evaluating the effect of ESI on the need for surgery as a primary outcome examined the same patient cohort, providing moderate evidence that patients who received ESI were less likely to undergo surgery than those who received control treatment. For studies examining surgery as a secondary outcome, ESI demonstrated a trend to reduce the need for surgery for short-term (<1 year) outcomes (risk ratio, 0.68; 95% confidence interval, 0.41–1.13; p=.14) but not long-term (≥1 year) outcomes (0.95, 0.77–1.19, p=.68). Secondary analyses provided low-level evidence suggesting that between one-third and half of patients considering surgery who undergo ESI can avoid surgery.

CONCLUSIONS: Epidural steroid injections may provide a small surgery-sparing effect in the short term compared with control injections and reduce the need for surgery in some patients who would otherwise proceed to surgery. © 2015 Elsevier Inc. All rights reserved.

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Introduction

Low back pain is the leading cause of disability globally in individuals under 45 years of age [1], with annual cost estimates upward of $100 billion in the United States [2]. Among the various procedural interventions for back pain, epidural steroid injections (ESI) are the most widely used, with the utilization rates more than doubling in the past 8 years [3]. Patients not responding to conservative treatments often undergo surgery. Similar to ESI, surgery rates have more than doubled over the past decade [4]. But despite the increase in back pain interventions, disability rates continue to rise [5,6]. Perhaps more concerning is a lack of significant correlation between the number of procedures performed and long-term outcomes [7,8]. As health-care costs consume a growing portion of the economy, the cost-effectiveness for medical interventions has taken on an increasingly prominent role.

There have been few cost analyses for ESI, with the ones that have been performed yielding mixed results [9–12]. Two ways to demonstrate whether ESI are cost-effective are to evaluate whether they facilitate return to work, as lost productivity accounts for over half of the economic costs of low back pain [13], or whether they prevent expensive treatments like surgery. In individuals unemployed secondary to low back pain, the likelihood of returning to work declines exponentially with the length of disability [14]. The likelihood of someone on disability for more than 1 year returning to work is so low that it is not considered to be a core outcome measure for chronic pain [15]. Using surgical intervention as a primary outcome measure of the cost-effectiveness of ESI is challenging, as few absolute indications for spine surgery exist, and subjective factors relating to patient and physician motivations assume principal roles in determining which patients receive surgery. Yet in many ways, the ability to prevent surgery should represent a key outcome measure for ESI, as it is objective (whereas pain is always subjective), reflects sustained and long-term treatment failure, and can dramatically alter cost-utility analyses. Among patients who present with absolute surgical indications (ie, cauda equina syndrome), ESI should not alter the treatment course. In the only randomized controlled trial (RCT) whose primary goal evaluated the ability of ESI to prevent surgery, Riew et al. [16] found that 29% of patients who received ESI underwent surgery at follow-ups ranging from 13 to 28 months versus 67% in a control group treated with epidural bupivacaine. At 5-year follow-up, most patients who did not undergo surgery at their first evaluation continued to avoid surgery [17]. However, secondary analyses of most RCTs have failed to demonstrate a surgery-sparing effect [9,10,18–34]. The purpose of this systematic review and meta-analysis is to determine whether ESI reduce the need for surgery, compared with control treatments.

Methods

Data sources and searches

This quantitative systematic review and meta-analysis followed the Cochrane and Preferred Reporting Items for Systematic Reviews and Meta-Analyses methodological guidelines [35,36]. Searches on Cochrane, PubMed, and EMBASE databases were performed without publication date or language restriction in January 2013. Search limitations included RCTs and subgroup analyses of RCTs. Additional studies were identified through hand searches of ESI review reference lists. Fig. 1 and Supplementary Table 1 illustrate further details of search strategy, including specific search terms.

Study selection

All authors performed the study selection. Three types of studies were included in this analysis. The main analysis (n=22) consisted of RCTs comparing ESI with a control group in which the reported outcomes included the need for surgery, with two studies reporting on the same cohort at different time points [16,17]. In a secondary analysis, subgroup analyses of RCTs comparing surgery with nonsurgical care were included, in which some patients received ESI, whereas others did not (n=2). In this group, the need for surgery was evaluated by the crossover rate for the surgical and nonsurgical groups. The final type of secondary analysis considered RCTs in which patients were allocated to either ESI or surgery (ie, all patients were surgical candidates and willing to have surgery), wherein the number of ESI to surgery crossovers was noted (n=2). Two studies in this category compared ESI with surgery but were excluded on the grounds that they failed to report crossover data [37,38]. Although the secondary analyses do not provide the same level of evidence as the main analysis, when placed in context, they may provide useful information regarding the ability of ESI to prevent surgery in a clinical, rather than controlled, setting [39].

Among 1,100 potentially eligible studies, 301 duplicate references were excluded, with another 742 deemed ineligible by title and abstract during initial screening, leaving 57 articles for consideration. After full manuscript review, 31 additional studies were excluded for failure to report surgical outcome (n=22), lack of ESI (n=5), lack of non-ESI control group (n=3), and lack of randomization (n=1), leaving 26 studies for systematic review and meta-analysis.

Outcome definition

To enhance generalization, a liberal definition of "need for surgery" included patients referred for surgery when no definitive disposition was provided. The rationale for this decision included consistency with the primary purpose

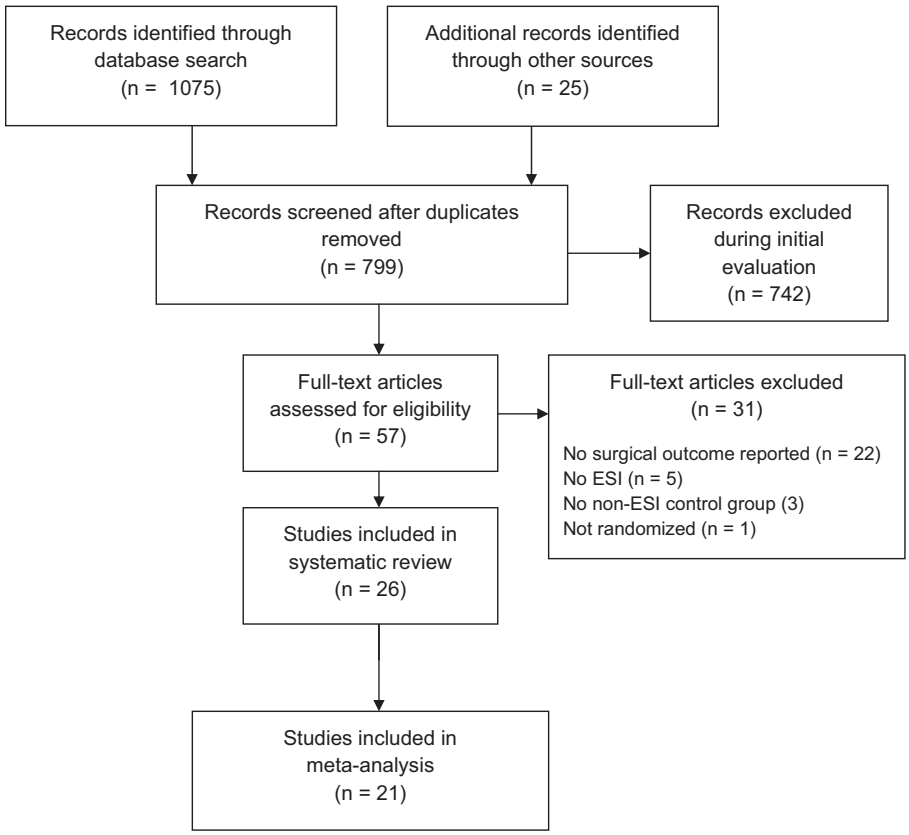

Fig. 1. Flow diagram demonstrating study search results. ESI, epidural steroid injection.

(ie, to determine whether ESI reduces the need for surgery, rather than surgery itself, which is subject to manifold patient and physician-related factors), lack of standardization as to what constitutes an appropriate surgical candidate, regional or era differences in surgical rates, variation in semantics and translation for studies performed in non–English-speaking countries, and minimization of inaccuracies in disposition (eg, patients listed as having surgery but who did not and vice versa), which may be especially relevant for older studies performed before electronic medical records were available. The need for surgery outcome was divided by time intervals into short-term (<1 year) and long-term (≥1 year) results.

Data extraction

Data extraction was performed independently by two authors (MCB and JTH) and included patient characteristics, control and treatment therapies, and need for surgery outcomes. Disagreements in data interpretation were resolved by a third author (SPC).

Quality and risk-of-bias assessments

Risk of bias was assessed using the methods recommended in the Cochrane Handbook for Systematic Reviews of Interventions [40], whereas a technical quality scale evaluated stringency of selection criteria for the RCTs comparing ESI with a sham injection [41]. All bias and technical quality ratings were performed by two authors independently (MCB and JTH). In the event of disparate ratings, a third author (SPC) adjudicated the results. Low methodological quality studies had at least one "high likelihood-of-bias" Cochrane risk domain, whereas low technical quality studies scored less than four points on the ESI technical quality scale.

The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach [42] was used to evaluate the overall quality of evidence using a version of the criteria from the Cochrane Back Review Group [35]. Quality of evidence was downgraded by one level for each of the factors we considered: design limitations (non-RCTs or >25% of participants from studies with low methodological quality); inconsistency (<75% of participants from studies with the findings in the same direction); indirectness (ie, heterogenous patient population); imprecision (<300 participants for each outcome); and reporting bias (funnel plot showing evidence of small study effects for RCTs or perception of reporting bias for non-RCTs). The presence of small study effects was assessed via visual inspection for funnel plot asymmetry and Egger test for all studies included in the meta-analysis. Quality of evidence would be downgraded for all meta-analyses by one level if the Egger test was statistically significant ($p<.100$). Two authors judged factors independently (MCB and JTH), with a third author adjudicating

disparate ratings. A GRADE assessment was completed for each pooled estimate. The following definitions of quality and evidence were used [43,44]: high quality (further research is very unlikely to change confidence in the estimate of effect), moderate quality (further research is likely to have an important impact on confidence in the estimate of effect and may change the estimate), low quality (further research is very likely to have an important impact on confidence in the estimate of effect and is likely to change the estimate), and very low quality (the estimate is very uncertain).

Data synthesis and statistical analysis

The principal summary measure for the primary analysis was the need for surgery (dichotomous). The effect size of this outcome was calculated as relative risk (RR), which represents the risk of needing surgery in the ESI treatment group divided by the risk of needing surgery in the control group. Analyses based on the outcome duration (short term <1 year and long term ≥1 year), quality and risk-of-bias assessments, and GRADE profile were performed.

For secondary analyses, qualitative and quantitative conclusions are presented in a descriptive manner. Because ESI tend to be more effective and surgical indications are more clear cut for neuropathic (particularly radicular pain stemming from a herniated disc) pain rather than mechanical pain [45–54], separate subgroup analyses were performed for RCTs comparing ESI with control injections in which only patients with neuropathic pain (eg, spinal stenosis and radiculopathy) were enrolled. Other subgroup analyses were performed based on the type of ESI performed (ie, caudal, transforaminal, or interlaminar technique) and in patients with no prior surgery, as both these variables have been shown to affect outcome [55–58].

Random-effect models were examined. Heterogeneity was measured by I^2, which assessed the variability among studies not attributable to chance alone. Significant heterogeneity was present with I^2 values more than 50%. Quality analysis was performed excluding low-quality studies for both methodological and technical scores. Calculations were done using Microsoft Excel 2011 (Microsoft Corp., Redmond, WA, USA) and RevMan (version 5.1.7; The Nordic Cochrane Center, The Cochrane Collaboration, 2011, Copenhagen, Denmark). Statistical significance for all tests was set at p value .05 or less.

Results

Systematic review

RCTs evaluating ESI versus control

The ability of ESI to prevent surgery was assessed by three different methods. In the first and primary analyses, we examined the 22 RCTs comparing ESI with a nonsteroid group in which the number of subjects who proceeded to surgery was reported [9,10,16–34,59]. Table 1 summarizes the study design, patient population, injection groups, outcome measures, and results. Thirteen studies were deemed to be of both high methodological and technical qualities (Supplementary Fig. 1, Table 2). Sample sizes ranged from 28 to 228 patients. Outcome variables included either "surgery" or a specific surgical procedure in 17 studies and "need for surgery" in 2 studies, whereas "proceeding to surgery," "transfer to surgery," and "referral to surgery" were present in 1 study each. In all five studies reporting short-term (<1 year) outcome data, no difference in surgery was found between ESI and non-ESI treatment groups [23,25–27,59]. Among the 17 studies reporting long-term (≥1 year) outcome data, no difference in surgery rates was found between ESI and comparison groups in most (15 out of 17, 88.2%) [9,10,18–22,26,28–34] but not all (2 out of 17, 11.8%) [16,17] studies. For all studies suggesting that ESI has no effect on need for surgery, the need for surgery was a secondary outcome measure.

Riew et al. [16,17] performed the only study evaluating the effect of ESI on need for surgery as a primary outcome. In this double-blind study in which surgical criteria were standardized and both surgeons and patients agreed surgery was the "next step," an orthopedic surgery group compared the surgical rate in patients with *herniated nucleus pulposus* or spinal stenosis who were randomized to receive either transforaminal ESI or epidural bupivacaine. At 13- to 28-month follow-up [16], fewer patients in the ESI treatment group underwent surgery (8 out of 28, 28.6%) compared with the control group (18 out of 27, 66.7%). Subsequent follow-up to 5 years [17] revealed that most patients who avoided surgery for the first year continued to avoid surgery at subsequent time periods. However, all eight patients lost to follow-up were in the treatment group, precluding definitive conclusions from being drawn. The overall quality of evidence for these effects was rated as moderate because of design imprecision.

Secondary analyses

Subgroup analyses of RCTs comparing surgery with nonsurgical treatment

In the second method, we examined two RCTs comparing surgery with nonsurgical treatment, in which the number of patients in the nonsurgical group who received ESI were compared with those who did not receive ESI, with regard to crossover to surgery [60,61]. Both studies were subgroup analyses of the Spine Patient Outcomes Research Trial, in which patients amenable to surgery with either lumbar radiculopathy secondary to a herniated disc or spinal stenosis were randomized by spine surgeons to either usual nonoperative care or decompression [62,63]. Both subgroup analyses retrospectively compared patients who did not receive ESI before study enrollment but were treated with ESI within 3 months of study enrollment with those who did not receive ESI.

The first subgroup analysis examined the rate of crossover in patients with lumbar disc herniation [60]. Among

Table 1
Characteristics of included studies

Author (year)	Study design	Patient population	Treatment group	Control group	Outcome measures	Results (patients needing surgery/total patients)	Comments
Randomized studies comparing ESI with nonepidural control							
Short term (<1 y) need for surgery							
Dilke et al. (1973) [23]	RCT P DB	100 patients with unilateral lumbosacral radiculopathy.	ESI with methylprednisolone 80 mg in 10-mL saline. Repeat injection at 1 wk if needed.	Ligament injection with 1-mL normal saline. Repeat injection at 1 wk if needed.	Pain relief, physical examination, analgesic consumption, work status, and surgery. Follow-up to 3 mo.	ESI, 7/52; non-ESI, 10/48.	Fluoroscopy not used. All patients received rehabilitation.
Hegihara et al. (2008) [25]	RCT P	69 patients with lumbar radiculopathy.	One TF ESI with betamethasone 4 mg and lidocaine.	One TF ESI with lidocaine.	Pain relief and surgery. Follow-up to 1 wk.	ESI, 3/34; non-ESI, 7/35.	Unknown baseline patient characteristics. Randomization by medical record number.
Klenerman et al. (1984) [26]	RCT P	63 patients with unilateral sciatica. Symptoms <6 mo in duration.	Single lumbar IL ESI with methylprednisolone 80 mg in 20-mL saline.	Single lumbar IL injection of 20-mL bupivacaine 0.25%. Single lumbar IL injection of 20-mL saline. Dry needling into the interspinous ligament.	VAS, physical examination, and surgery. Follow-up to 2 mo.	ESI, 0/19; non-ESI, 2/44.	Fluoroscopy not used.
Kraemer et al. (1997) [27]	RCT P	133 patients with unilateral lumbosacral radiculopathy from single nerve root compression.	Three TF ESI in 1 wk with triamcinolone 10 mg with 1-mL local anesthetic. Three IL ESI of same injectate.	Three paravertebral injections in 1 wk with local anesthetic.	Pain relief, work status, physical examination, and surgery. Follow-up to 3 mo.	ESI, 11/87; non-ESI, 6/46.	Fluoroscopy not used.
Laiq et al. (2009) [59]	RCT P	52 patients with lumbar radiculopathy. Symptoms >2 wk in duration.	One IL ESI with methylprednisolone 80 mg, 3-mL lidocaine 2%, and 5-mL normal saline, preceded by negative test dose of 3-mL lidocaine 2%.	Bed rest, nonsteroidal anti-inflammatory medication, muscle relaxants, and opioids.	VAS, satisfaction, and surgery. Follow-up to 6 mo.	ESI, 4/26; non-ESI, 6/26.	Fluoroscopy not used. One control patient and one treatment group patient lost to follow-up.
Long term (≥1) need for surgery							
Arden et al. (2005) [18]	RCT P DB	228 patients with a clinical diagnosis of unilateral lumbar radiculopathy. Acute symptoms in one-third of patients.	Up to 3 IL ESI at 3-wk intervals with triamcinolone 80 mg and 10 mL of bupivacaine 0.25%.	Up to three IS ligament injections at 3-wk intervals with 2 mL of normal saline.	VAS, ODI, missed work, analgesic usage, and surgery. Follow-up to 1 y.	ESI, 17/113; non-ESI, 13/95.	Fluoroscopy not used. MRI not used to confirm pathology.

Bush and Hillier (1991) [19]	RCT P DB	28 patients with lumbosacral radiculopathy.	Two caudal injections at a 2-wk interval with triamcinolone 80 mg and 25 mL of procaine 0.5%.	Two caudal injections at a 2-wk interval with 25 mL of normal saline.	VAS, QOL, physical examination, and surgery. Follow-up to 1 y.	ESI, 1/13; non-ESI, 2/15.	Fluoroscopy not used. Of five patient "dropouts" because of worsening symptoms, three required surgery.
Carette et al. (1997) [20]	RCT P DB	158 patients with lumbosacral radiculopathy from herniated disc. Symptoms <12 mo in duration.	Up to three ESI at 3-wk intervals with 80-mg methylprednisolone in 8-mL normal saline.	Up to three epidural injections at 3-wk intervals with 1-mL normal saline.	VAS, ODI, physical examination, analgesic consumption, and surgery. Follow-up for surgery to 1 y, other variables 3 mo.	ESI, 20/78; non-ESI, 20/80.	Fluoroscopy not used.
Cohen et al. (2012) [21]	RCT P DB	84 patients with lumbosacral radiculopathy from disc pathology. Symptoms <6 mo in duration.	Two TF injections at 2-wk intervals with methyl-predniosolone 60 mg in 0.5-mL bupivacaine 0.5%. Two TF injections at 2-wk intervals of etanercept 4 mg with 0.5-mL bupivacaine 0.5%.	Two TF injections at 2-wk intervals of 0.5-mL bupivacaine 0.5% with 1.5-mL normal saline.	VAS, functional capacity, medication usage, and surgery. Follow-up for surgery to 1 y, other variables 6 mo.	ESI, 12/54; non-ESI, 5/30.	Treatment failures unblinded at 1 mo.
Cuckler et al. (1985) [22]	RCT P DB	73 patients with acute lumbar herniated disc or spinal stenosis.	One ESI at L3–L4 with methylprednisolone 80 mg in 5 mL of procaine 1% plus 2-mL water.	One epidural injection at L3–L4 with 5-mL procaine 1% and 2-mL saline. If <50% improvement within 24 h then treatment injection given.	Pain relief >75%, surgery. Follow-up to 1 y.	ESI, 16/42; non-ESI, 7/31.	Fluoroscopy not used. At 24 h, 36 patients (18 in each group) with <50% improvement had a nonblinded ESI treatment injection. Steroids take >24 h to exert full effect. Follow-up ranged from 13 to 30 mo.
Ghahreman et al. (2010) [24]	RCT P DB	130 patients with lumbosacral radiculopathy from herniated disc.	One TF ESI with triamcinolone 70 mg and 0.75-mL bupivacaine 0.5%. Repeat injection at 1 wk if needed.	One TF injection with 2-mL bupivacaine 0.5%. Single TF injection with 2-mL normal saline. Intramuscular injection triamcinolone 70 mg (1.75 mL). Intramuscular injection with 2-mL normal saline. Repeat injection at 1 wk if needed.	Pain relief, functional and psychological improvement, and surgery. Follow-up for surgery to 1 y, other outcomes to 1 mo.	ESI, 10/28; non-ESI, 31/122.	Lower percentage of treatment group had chronic pain.

(*Continued*)

Table 1
(*Continued*)

Author (year)	Study design	Patient population	Treatment group	Control group	Outcome measures	Results (patients needing surgery/total patients)	Comments
Iversen et al. (2011) [33]	RCT P DB	116 patients with lumbar radiculopathy. Symptoms >12 wk in duration.	Two caudal ESI at 2-wk interval with triamcinolone 40 mg+29-mL saline.	Two subcutaneous injections at 2-wk intervals with 2-mL saline. Two caudal injections at 2-wk intervals with 30-mL saline.	VAS, ODI, QOL, and surgery. Follow-up to 1 y.	ESI, 1/37; non-ESI, 14/79.	Sham group had greater baseline disease burden. One-milliliter steroid diluted in 30 mL is extremely small dose for caudal injection.
Karppinen et al. (2001) [10]	RCT P DB	160 patients with unilateral lumbosacral radiculopathy. Symptoms <6 mo in duration.	Single lumbar TF ESI with 40-mg methylprednisolone 40-mg in 2-mL bupivacaine 0.5%. Single S1 TF ESI with methylprednisolone 40 mg in 3-mL bupivacaine 0.75%.	Same TF injection scheme with normal saline constituting same volume.	VAS, ODI, QOL, physical examination, and surgery. Follow-up to 1 y.	ESI, 18/80; non-ESI, 15/80.	
Mathews et al. (1987) [28]	RCT P DB	57 patients with uniradicular lumbar pain with neurologic deficit.	Up to three caudal ESI at 2-wk intervals with 2-mL methylprednisolone 80 mg and 20-mL bupivacaine 0.125%.	Up to three intermuscular injections with 2-mL lidocaine over sacral hiatus or most tender spot.	Pain scores, medication usage, and surgery. Follow-up to 1 y.	ESI, 1/23; non-ESI, 0/34.	Fluoroscopy not used.
Price et al. (2005) [9]	RCT P DB	228 patients with unilateral lumbosacral radiculopathy. Symptoms <18 mo in duration.	Up to three ESI at 3-wk intervals with triamcinolone 80 mg and 10-mL bupivacaine 0.125%.	Up to three injections at 3-wk intervals with 2-mL normal saline into interspinous ligament.	VAS, ODI, physical examination, QOL, psychological status, and surgery. Follow-up to 1 y.	ESI, 15/120; non-ESI, 14/108.	Fluoroscopy not used. Randomization stratified by pain duration.
Riew et al. (2000) [16] and Riew et al. (2006) [17]	RCT P DB	55 patients with lumbar radiculopathy from herniated disc or spinal stenosis.	Up to four ESI at random interval with betamethasone 6 mg and 1-mL bupivacaine 0.25%.	Up to four ESI at random interval with 1-mL bupivacaine 0.25%.	North American Spine Society outcome instrument and surgery. Follow-up to 5 y.	ESI, 8/28; non-ESI, 18/27. At 13- to 28-mo follow-up, need for surgery was significantly less in treatment (29%) than in control group (67%). At 5 y, difference became nonsignificant because of lost follow-ups.	Need for surgery was primary outcome. All patients were candidates for surgery. All 8 patients lost to 5-y follow-up were in treatment group.
Sayegh et al. (2009) [34]	RCT P DB	183 patients with low back pain with or without radiculopathy. Symptoms >1 mo in duration.	Up to two caudal ESI at 1 mo intervals with 1-mL betamethasone and 12-mL lidocaine 2%.	Up to two caudal epidural injections at 1-mo intervals with 12-mL lidocaine 2% and 8-mL sterile water.	ODI, physical examination, and surgery. Follow-up to 1 y.	ESI, 13/93; non-ESI, 19/90.	Fluoroscopy not used.

Snoek et al. (1977) [29]	RCT P DB	51 patients with unilateral lumbosacral radiculopathy from herniated disc.	Single ESI with methylprednisolone 80 mg in 2-mL saline.	Single epidural injection with 2-mL normal saline.	Pain relief, physical examination, analgesic consumption, and surgery. Follow-up to 8–20 mo.	ESI, 14/27; non-ESI, 14/24.	Fluoroscopy not used.
Tafazal et al. (2009) [30]	RCT P DB	50 patients with unilateral lumbosacral radiculopathy from herniated disc or spinal stenosis.	Single TF ESI with methylprednisolone 40 mg and 2-mL bupivacaine 0.25%.	Single TF epidural injection with 2-mL bupivacaine 0.25%.	VAS, ODI, and surgery. Follow-up to 1 y.	ESI, 9/55; non-ESI, 14/51.	Follow-up ranged from 12 to 31 mo.
Valat et al. (2003) [31]	RCT P DB	85 patients with lumbosacral radiculopathy from herniated disc.	Three ESI at 2 d intervals with methylprednisolone 50 mg in 2 mL.	Three epidural injections with 2-mL normal saline.	VAS, physical examination, functional capacity, and surgery. Follow-up to 35 d.	ESI, 1/42; non-ESI, 2/42.	Fluoroscopy not used. No lumbar exercises allowed.
Wilson-MacDonald et al. (2005) [32]	RCT P DB	93 patients with lumbosacral radiculopathy from herniated disc and/or spinal stenosis. Symptoms >6 wk in duration.	Up to two IL ESI at random interval with 2-mL methylprednisolone 80 mg and 8-mL bupivacaine 0.5%.	Single intramuscular injection with 2-mL methylprednisolone 80 mg and 8-mL bupivacaine 0.5%.	Pain relief, ODI, and surgery. Follow-up to 2 y.	ESI, 18/44; non-ESI, 15/48.	All patients were candidates for surgery. Fluoroscopy not used. 15% drop-out rate.
Subgroup analyses of randomized studies comparing surgery with conservative treatment							
Radcliff et al. (2012) [60]	Retrospective subgroup analyses of P RCT and observational study comparing surgery with conservative care in patients with herniated disc.	607 patients with radicular pain who had not received ESI before study enrollment. Symptoms >6 wk in duration.	ESI via TF, IL, or caudal route during first 3 mo of study enrollment, of variable dose and volume.	No ESI.	Crossover to undergo nonsurgical or surgical treatment. Follow-up to 4 y.	Patients who crossed over from surgery to no surgery within 3 mo of enrollment: ESI, 24/59 (41%); non-ESI, 37/304 (12%). Proportion of patients expressing preference for nonsurgical treatment: ESI, 139/453 (31%); non-ESI, 86/154 (56%).	Included both randomized and observational cohorts. Use of fluoroscopy not ascertained. Did not distinguish between type and number of ESI.
Radcliff et al. (2013) [61]	Retrospective subgroup analyses of P RCT and observational study comparing surgery with conservative care in patients with spinal stenosis.	278 patients with lumbar spinal stenosis who had not received ESI before study enrollment. Symptoms >6 wk in duration.	ESI via TF, IL, or caudal route during first 3 mo of study enrollment, of variable dose and volume.	No ESI	Crossover to undergo nonsurgical or surgical treatment. Follow-up to 4 y.	Patients who crossed over from surgery to no surgery within 3 mo of enrollment: ESI: 7/21 (33%); non-ESI, 13/122 (11%). Patients who crossed over from nonsurgical treatment to surgery within 3 m of enrollment: ESI, 28/48 (58%); non-ESI, 27/85 (32%).	Included both randomized and observational cohort. Use of fluoroscopy not ascertained. Did not distinguish between type and number of ESI. Trend for worse stenosis at baseline in ESI group.

(*Continued*)

Table 1
(*Continued*)

Author (year)	Study design	Patient population	Treatment group	Control group	Outcome measures	Results (patients needing surgery/total patients)	Comments
						Proportion of patients expressing preference for nonsurgical treatment: ESI, 43/69 (62%); non-ESI, 68/207 (33%). ESI treatment increased crossover to nonsurgical treatment in those allocated to surgery compared with control group. Epidural spine injection increased crossover to surgery compared with control group in those allocated to nonsurgical care. Higher proportion of patients who received ESI preferred nonsurgical treatment.	
Randomized studies comparing ESI with surgery							
Butterman (2004) [64]	RCT P	100 patients with herniated nucleus pulposus. Symptoms >3 y.	Up to three ESI with betamethasone 10 or 15 mg.	Discectomy.	Crossover to undergo surgical treatment. Follow-up to 2–3 y.	Among 50 patients in the ESI group, 27 (46%) did not have surgery.	Variable use of fluoroscopy.
Gerszten et al. (2010) [65]	RCT P	90 patients with herniated nucleus pulposus. Symptoms >1 mo.	Up to two TF ESI of variable dose and volume at 3-wk intervals.	Plasma disc decompression.	Crossover to undergo surgical treatment. Follow-up to 2 y.	Among 40 patients in the ESI group, 50% did not receive surgery.	Use of fluoroscopy not ascertained. All patients had already failed ESI.

DB, double blind; ESI, epidural steroid injection; IL, interlaminar; IS, Interspinous; MRI, magnetic resonance imaging; ODI, Oswestry disability index; P, prospective; QOL, quality of life; RCT, randomized controlled trial; TF, transforaminal; VAS, visual analog scale.

Table 2
Risk of bias and technical quality assessment of included studies

Study name	Qualitative bias assessment						ESI technical quality score
	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	
ESI vs. nonepidural control							
Short term (<1 y) need for surgery							
Dilke et al. (1973) [23]	+	?	?	?	+	+	5
Hegihara e al. (2008) [25]	−	−	?	?	+	+	3
Klenerman et al. (1984) [26]	+	?	−	−	+	?	4
Kraemer et al. (1997) [27]	?	?	−	−	+	?	5
Laiq et al. (2009) [59]	?	?	−	−	+	?	2
Long-term (≥1 y) need for surgery							
Arden et al. (2005) [18]	+	+	+	+	+	+	7
Bush and Hillier (1991) [19]	?	?	+	+	+	+	4
Carette et al. (1997) [20]	+	+	+	+	+	+	8
Cohen et al. (2012) [21]	+	+	+	+	+	+	11
Cuckler et al. (1985) [22]	?	?	?	?	+	?	5
Ghahreman et al. (2010) [24]	?	?	+	+	+	+	8
Iversen et al. (2011) [33]	?	+	−	−	+	+	6
Karppinen et al. (2001) [10]	+	+	+	+	+	+	9
Mathews et al. (1987) [28]	+	?	+	+	?	+	4
Price et al. (2005) [9]	+	+	+	+	+	+	7
Riew et al. (2000) [16] and Riew et al. (2006) [17]	?	+	+	+	+	+	6
Sayegh et al. (2009) [34]	?	?	+	+	?	−	4
Snoek et al. (1977) [29]	?	?	+	+	?	?	7
Tafazal et al. (2009) [30]	+	?	+	+	+	+	7
Valat et al. (2003) [31]	+	+	+	+	+	+	10
Wilson-MacDonald et al. (2005) [32]	+	+	+	+	+	−	3
Nonsurgical ESI vs. nonsurgical control							
Radcliff et al. (2012) [60]	+	+	−	+	+	+	3
Radcliff et al. (2013) [61]	+	+	−	+	+	+	3
ESI vs. surgery							
Butterman (2004) [64]	+	?	−	−	+	+	4
Gerszten et al. (2010) [65]	+	+	−	−	+	+	7

ESI, epidural steroid injection.

patients randomized to surgery, more patients who received ESI within 3 months of enrollment expressed a preference for nonsurgical treatment (ESI, 56%; no ESI, 31%; p<.001). More patients who received ESI crossed over to nonsurgical management compared with patients who did not receive ESI (ESI, 41%; no ESI, 12%; p<.001). In the same study, among patients randomized to nonsurgical treatment, more patients who received ESI crossed over to surgery compared with patients who did not receive ESI, although this trend was not significant (ESI, 42%; no ESI, 30%; p=.057). The second subgroup analysis of patients with lumbar spinal stenosis yielded similar results [61]. More patients who received ESI within 3 months of enrollment expressed a preference for nonsurgical treatment (ESI, 62%; no ESI, 33%; p<.001), and a higher rate of crossover from surgery to nonsurgical treatment was observed for ESI patients (ESI, 33%; no ESI, 11%; p=.012). However, in the nonsurgical group, more patients receiving ESI crossed over to surgery (ESI, 58%; no ESI, 32%; p=.003). When data from both studies were combined, more patients randomized to surgery who received ESI crossed over to nonsurgical treatment (ESI, 39%; no ESI, 12%; p<.001). However, for patients randomized to no surgery, a higher proportion who received ESI crossed over to surgery (ESI, 49%; no ESI, 16%; p≤.001). The overall quality of evidence for these effects was rated as low because of design limitations and imprecision.

Randomized trials comparing ESI with surgery

The third way we sought to examine whether ESI might decrease the likelihood of surgery was to evaluate RCTs comparing ESI with surgery in patients referred for surgery, wherein crossover rates were reported [64,65]. In a comparative-effectiveness study comparing ESI with discectomy in 50 patients with a large single-level herniated disc who were referred for surgical evaluation, Butterman [64] found that 46% of patients treated with a series of ESI did not receive surgery at the 2- to 3-year follow-up period. In another comparative-effectiveness study comparing ESI with plasma disc decompression in 85 patients with sciatica from a single-level–contained disc protrusion, Gerszten et al. [65] found a similar proportion of patients

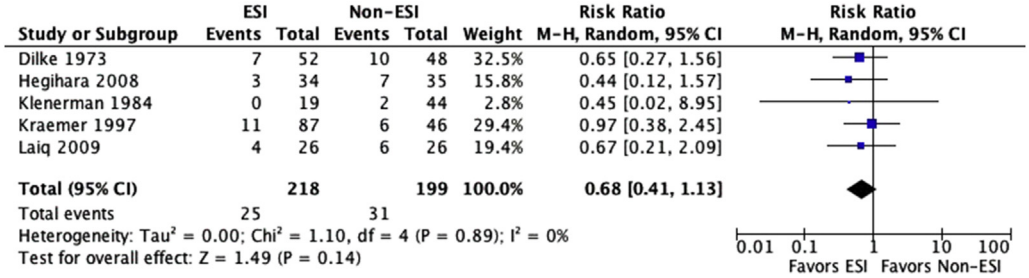

Fig. 2. Effect of epidural steroid injection (ESI) on need for surgery in the short-term (<1 year) outcome. Diamond indicates the risk ratio. The overall summary estimate for the analysis (width of the diamond represents 95% confidence interval [CI]). Boxes indicate the weight of the individual studies in the pooled analysis. Whiskers indicate the 95% CIs. Midline represents risk ratio of 1. When to left of midline, the risk of needing surgery favors (ie, is less with) ESI. When to right of midline, the risk of needing surgery favors (ie, is less with) non-ESI. If lines touch midline, the risk of needing surgery is comparable for the two injection arms. M-H, Mantel-Haenszel.

allocated to receive ESI did not undergo surgery (50%) at 2-year follow-up. Of note, all patients in this study had already received previous ESI. The overall quality of evidence for these effects was rated as low because of design limitations and imprecision.

Meta-analysis

Primary analysis

In the analysis of need for surgery in the short-term (<1 year), 417 patients from 5 studies provided data (Fig. 2). A lower percentage of patients in the ESI group demonstrated a need for surgery (11.5% [25 out of 218]) compared with the non-ESI group (15.6% [31 out of 199]). Epidural spine injection demonstrated a trend to reduce the need for surgery in the short term that fell shy of statistical significance (RR, 0.68; 95% confidence interval [CI], 0.41–1.13; I^2=0%; p=.14). When studies of low methodological or technical quality were excluded [25–27,59], the estimate of reduction in need for surgery was slightly less convincing for the one remaining study of high quality (RR, 0.65; 95% CI, 0.27–1.56; p=.33) [23]. Inspection of the funnel plot and lack of statistical significance for the Egger test (p=.169) suggested no small study effects (Supplementary Fig. 2); therefore, no meta-analysis had its quality of evidence downgraded because of reporting bias. The overall quality of evidence for the short-term effect of ESI on need for surgery was rated as moderate because of design limitations.

Long-term (≥1 year) outcome data on need for surgery were provided from 1,834 patients in 16 studies (Fig. 3). For the two studies reporting on the same cohort at different long-term time points [16,17], data from the longest duration of follow-up were used [17]. Although fewer patients in the ESI group (19.8% [174 out of 878]) demonstrated a need for surgery compared with the nonsteroid group (21.2% [203 out of 956]), ESI did not significantly affect the need for surgery in the long term (RR, 0.95; 95% CI,

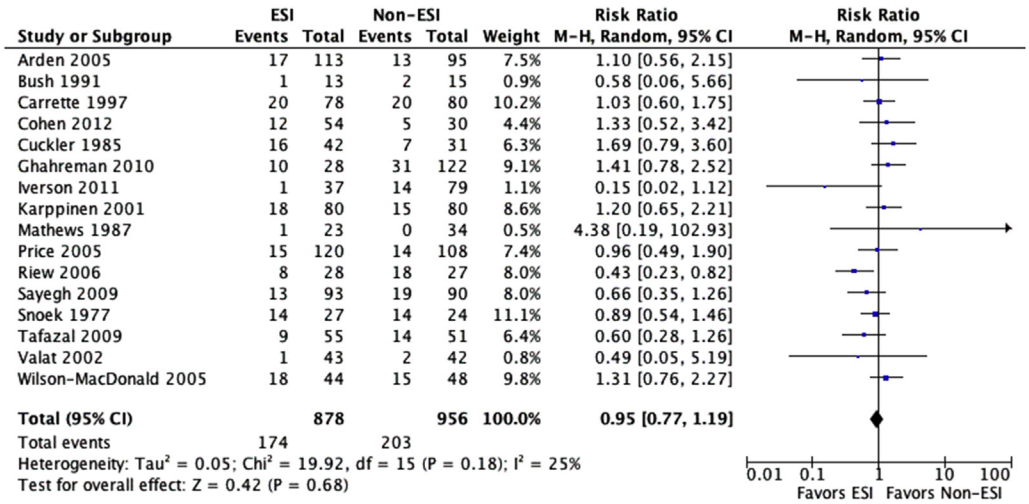

Fig. 3. Effect of epidural steroid injection (ESI) on need for surgery in the long-term (≥1 year) outcome. Diamond indicates the risk ratio. The overall summary estimate for the analysis (width of the diamond represents 95% confidence interval [CI]). Boxes indicate the weight of individual studies in the pooled analysis. Whiskers indicate the 95% CIs. Midline represents risk ratio of 1. When to left of midline, the risk of needing surgery favors (ie, is less with) ESI. When to right of midline, the risk of needing surgery favors (ie, is less with) non-ESI. If lines touch midline, the risk of needing surgery is comparable for the two injection arms. M-H, Mantel-Haenszel.

0.77–1.19; I^2=25%; p=.68). When studies of low methodological or technical quality were excluded [32,33], the estimate of reduction in need for surgery did not change (RR, 0.94; 95% CI, 0.76–1.17; I^2=15%; p=.59). The overall quality of evidence for the long-term effect of ESI to reduce need for surgery was rated as moderate because of inconsistency.

A combination of both short- and long-term outcomes resulted in the analysis of 2,251 patients from the 22 studies [9,10,16–34,59]. Although fewer patients demonstrated a need for surgery in the ESI group (18.2% [199 out of 1,096]) in comparison with the non-ESI group (20.3% [234 out of 1,155]), the difference did not reach statistical significance for the entire cohort (RR, 0.92; 95% CI, 0.76–1.11; I^2=12%; p=.37) or when those studies of low quality were excluded (RR, 0.93; 95% CI, 0.76–1.14; I^2=12%; p=.46). The quality of evidence for the overall effect was rated as moderate because of inconsistency.

Secondary analyses

For all subgroup analyses, no differences were found for need for surgery in patients receiving ESI compared with control treatment. Epidural spine injection did not affect need for surgery among either patients with neuropathic back pain (ie, *herniated nucleus pulposus*, spinal stenosis, or foraminal narrowing) [19–21,23,26,28–31] (22.9% [101 out of 442] vs. 25.8% [117 out of 453]; RR, 0.93; 95% CI, 0.69–1.26; I^2=35%; p=.65) or among patients with no history of surgery [9,14,16,17,19–23,27–29,31,59] (17.1% [131 out of 766] vs. 19.3% [167 out of 864]; RR, 0.94; 95% CI, 0.77–1.16; I^2=0%; p=.56). Surgical rates also did not differ when ESI were stratified by type: caudal (9.6% [16 out of 166] vs. 16.1% [35 out of 218]; RR, 0.59, 95% CI, 0.26–1.34; I^2=15%; p=.21) [19,28,33,34]; transforaminal (20.7% [52 out of 251] vs. 22.6% [72 out of 318]; RR, 1.01, 95% CI, 0.68–1.51; I^2=27%; p=.96) [10,21,24,30], and interlaminar (20.5% [105 out of 512] vs. 18.7% [93 out of 498]; RR, 1.06; 95% CI, 0.84–1.34; I^2=0%; p=.64) [9,18,20,22,26,29,31,32,56]. For all subgroup analyses, stratification by outcome duration (short term vs. long term) and quality did not alter the results (data not shown).

Discussion

The findings in this systematic review and meta-analysis provide mixed results regarding the ability of ESI to reduce the need for surgery. The only study examining the impact of ESI on surgery as a primary outcome demonstrated moderate evidence for a surgery-sparing effect for ESI in the short term. Studies including surgery as a secondary outcome provide moderate evidence of no surgery-sparing effect: only a trend for short-term but not long-term benefit was found. Randomized controlled studies comparing surgery with ESI and non-ESI conservative care provide low-quality evidence to support ESI reducing the need for surgery, between one-third and one-half of those subjects who would otherwise likely have undergone surgery being able to avoid this intervention [60,61,64,65].

The main finding in this study is that there may be a weak surgery-sparing effect for ESI in the short term but not the long term. Demonstrating a short-term surgery-sparing effect is significant given the use of a liberal definition of need for surgery in this analysis, which reduces the likelihood of finding a correlation compared with more restrictive definitions of surgery. The short-term surgery-sparing effect of ESI complements the existing literature, as ESI have been shown to provide benefit for patients with radicular pain when other outcomes besides surgery are measured in the short and intermediate term [21,24,56]. Several reasons may help explain the discrepancy in short- and long-term outcomes. The most obvious one is that the beneficial effects of ESI are transient. The long-term effectiveness of ESI is limited because of either disease progression of the spine or of the duration of action of the steroid. Furthermore, patients and physicians may prefer a long-term solution compared with repeated ESI procedures. A second possible explanation is that in most controlled studies, ESI were not routinely repeated on an ‘‘as-needed’’ basis, as is often done in clinical practice. In one review, Roberts et al. [66] found that RCTs that allowed for multiple injections were more likely to report positive outcomes than studies that limited the number of injections to one.

The evaluation of need for surgery as a secondary outcome in most RCTs limits the ability to draw robust conclusions on the lack of a strong association between ESI and a reduced need for surgery. In contrast with the studies in which need for surgery is the primary outcome [16,17], studies that include surgery as a secondary outcome [9,10,18–34,59] fail to ensure that study participants are surgical candidates, are willing to even consider surgery, or that operative criteria are standardized. These limitations reduce the number of patients eligible for surgery and result in considerable variability in what should ideally be a standardized outcome measure, which in turn lower the power to detect a difference between groups. If ESI do possess a surgery-sparing effect, this effect is diluted by methodological flaws in studies that seek to evaluate this outcome as a secondary measure, and can perhaps only be overcome in studies performed by surgeons, with predetermined operative standards. Randomized controlled studies performed by surgeons and other nonpain specialists are almost three times as likely to yield a negative result than those performed by interventional pain physicians [55], suggesting the possibility of researcher bias. Despite this possible confounding factor, nearly all studies suggesting that ESI may prevent surgery involved spine surgeons [16,17,60,61,64].

Subgroup analyses regarding rates of crossover should be interpreted with caution. Because all patients enrolled in randomized surgical trials are ostensibly good candidates

Table 3
Recommendations for future study design to evaluate the ability of ESI to prevent surgery

- Involve multiple study sites and power the study to detect a relatively infrequent event
- Ensure all patients are surgical candidates
- Ensure all pathologies are amenable to surgery and that surgical indications are standardized
- Measure surgery as a primary outcome
- Randomize patients to a standardized technique for ESI or suballocate patients with different pathology who receive different types of injections (ie, unilateral pain to a transforaminal subgroup and bilateral pain to an interlaminar group)

ESI, epidural steroid injection.

amenable to surgery, when patients fail ESI—arguably the most studied invasive and expeditious "nonsurgical intervention"—they may be expected to crossover to surgery. The assertion that ESI may provide a small surgery-sparing effect to some patients allocated to surgical intervention is supported by the fact that higher percentages of patients crossed over to nonsurgical treatment after receiving ESI compared with those subjects who did not undergo ESI in both subgroup analyses [60,61] and that larger proportions of patients who received ESI expressed a preference for nonsurgical management.

This analysis examined ESI as an isolated treatment to prevent spine surgery, but common clinical practice employs ESI as a treatment used concomitantly with other conservative management options such as physical, pharmacologic, and complementary and alternative medical therapies. Future studies examining outcomes resulting from ESI used within a multimodal treatment model before surgery may further refine the selection of surgical candidates. The overall methodologic and technical qualities of many studies we evaluated were poor. Most offered limited information on the effectiveness of blinding [25–27,60], and blinding was not possible for patients randomized to surgery versus nonsurgical treatment [60,61,64,65]. To address this shortcoming, outcomes were compared between treatment and control groups in several subgroup analyses to include the exclusion of low-quality studies, with no difference in the findings. Second, many of the studies used different indications (ie, herniated disc vs. spinal stenosis), doses of steroid, volumes of injectate, and number of allotted injections. Although variations in the technical factors may affect outcomes, evidence-based analyses suggest that these factors have less of an effect on outcome than patient-related factors, such as psychosocial pathology [55,67]. Finally, the clinical utility and safety of both ESI [55,67–69] and surgery [54,70] for low back pain continue to be debated. With regard to ESI, systematic and evidence-based reviews are mixed regarding their effectiveness, [55,67,68] and it is well known that spinal injections may contain significant risks [69,71].

Conclusions

Epidural steroid injections may provide a small surgery-sparing effect in the short term (<1 year) compared with control injections, based on the only study examining need for surgery as the primary outcome and the trend of all the RCTs. Larger multicenter studies using standardized operative criteria in patients with surgically amenable pathology, and surgery as the primary outcome measure, are needed to better address this question (Table 3).

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Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.spinee.2014.10.011.

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